Automating with Bash

NUnix · L3 MIAGE · CM3 · 90 minutes

ISTIC · University of Rennes 2026–2027



Louis Ledoux

Writing Bash scripts

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A support team has several log files. We will build a script that counts matching lines and saves a report.

1. Scripts, arguments and variables
2. Tests and decisions
3. Loops and counters
4. Functions
5. Build a log report

1. Scripts, arguments and variables

How can a script use a filename we type or pass as an argument?

A first script

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Save these commands in `hello.sh`:

```
#!/bin/bash
# A first script
echo "Hello, MIAGE"
```

- `#!/bin/bash` selects Bash for direct execution: the **shebang**.
- `#` begins a comment in this position.
- Commands run in order. A newline or `;` separates them.

Two ways to run a script

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Invoke Bash

```
bash hello.sh
```

Bash reads the file.
It needs **read** access to it.

Execute the file

```
chmod u+x hello.sh
./hello.sh
```

Requires **execute** permission and a usable interpreter.

Assigning and reading a variable

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```
filename="app.log"
echo "$filename"
filename="worker.log"
echo "Selected file: $filename"
```

`filename=value` stores a value. **No spaces around =.**
`$filename` reads that value. Quote it: `"$filename"`.

Asking the user with read

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```
read -r -p "File to inspect: " filename
echo "Selected file: $filename"
```

Type `app.log`, then press **Enter**. The message becomes `Selected file: app.log`.

- `read` stores the typed line in `filename`.
- `-p` supplies the prompt displayed before the input.
- `-r` keeps backslashes as ordinary characters.

Positional parameters

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A script uses these parameters to read its arguments:

```
bash arguments.sh "L3 MIAGE" Unix
```

Parameter	Meaning	Value for this call
`$0`	Script name	`arguments.sh`
`$1`	First argument	`L3 MIAGE`
`$2`	Second argument	`Unix`
`$#`	Number of arguments	2
`"$@"`	All arguments separately	`"L3 MIAGE" "Unix"`
`"$*"`	All arguments as one string	`"L3 MIAGE Unix"`

Passing arguments with spaces

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Use the **same script call**:

```
bash arguments.sh "L3 MIAGE" Unix
```

Inside the script, `printf` uses `%s` for a string and `\n` for a newline.
It repeats the format for each argument.

"$@": two arguments

```
printf '<%s>\n' "$@"
```

```
<L3 MIAGE>
<Unix>
```

"$*": one string

```
printf '<%s>\n' "$*"
```

```
<L3 MIAGE Unix>
```

Use "$@" to pass each argument on unchanged.

Saving a command's output

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From the extracted TP files:

```
cd ~/NUnix/files
tpfiles=$(pwd)
cd ..
ls "$tpfiles"
```

Bash runs `pwd` and stores its printed path in `tpfiles`.
That **absolute path** still works after `cd ...`

Running commands in another Bash

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```
bash -c 'echo "Hello from another Bash"'
```

- `-c` tells Bash to execute the **following text as commands**.
- The current shell starts a **child Bash**, then waits for it to finish.
- Single quotes pass the text unchanged for that child to interpret.

`bash script.sh` reads commands from a file.
`bash -c 'commands'` reads them from the next argument.

Sharing a variable with a script

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```
tpfiles="$HOME/NUnix/files"
export tpfiles
bash -c 'echo "$tpfiles"'
```

- The assignment stores the path in the **current shell**.
- `export` includes it in the environment passed to newly started programs.
- The child Bash can now print the same path.

A child receives a **copy**. Changing it does not change the parent's variable.

2. Tests and decisions

Before reading a file, how can a script check that it exists and is readable?

Testing a condition

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Does the directory `logs` exist? These two commands ask the same question:

```
test -d logs
[ -d logs ]
```

`-d` means **is a directory**. `[ ... ]` is another way to write `test`.

A test returns a status; it prints no answer.
`0`: the condition is true. `1`: the condition is false.

Comparing integers

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Compare **integers** with these operators inside `[ ... ]`:

Operator	Read it as	Mathematical symbol
`-eq`	Equal to	=
`-ne`	Not equal to	≠
`-lt`	Less than	<
`-le`	Less than or equal to	≤
`-gt`	Greater than	>
`-ge`	Greater than or equal to	≥

Read `[ "$count" -gt 10 ]` as **count is greater than 10**.

Testing text

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An interactive script can stop when the user types end.

Test	True when...
`[ "$answer" = "end" ]`	The answer is end
`[ "$answer" != "end" ]`	The answer is different from end
`[ -z "$answer" ]`	The answer is empty
`[ -n "$answer" ]`	The answer is not empty

Use = for text and `-eq` for integer comparisons.

Testing files and permissions

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A script can check a path before using it:

Test	True when...
`[ -e "$filename" ]`	The path exists
`[ -f "$filename" ]`	It is a regular file
`[ -d "$filename" ]`	It is a directory
`[ -r "$filename" ]`	Reading is allowed
`[ -w "$filename" ]`	Writing is allowed
`[ -x "$filename" ]`	Executing / directory traversal is allowed

if and else

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```
count=12
if [ "$count" -gt 10 ]; then
    echo "More than ten"
else
    echo "Ten or fewer"
fi
```

A true test runs the `then` branch; a false test runs `else`. `fi` marks the end.

Spaces are required after `[` and before `]`.
Correct: `[ "$count" -gt 10 ]`
Incorrect: `["$count" -gt 10]`

More than two cases: elif

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```
count=10
if [ "$count" -gt 10 ]; then
    echo "More than ten"
elif [ "$count" -eq 10 ]; then
    echo "Exactly ten"
else
    echo "Fewer than ten"
fi
```

`elif` means **else if**. Try each test in order and run the first true branch.
Here, the output is `Exactly ten`.

Conditional command lists

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```
mkdir results && echo "Directory created"
test -d results || echo "Directory is missing"
```

- a && b: run b only when a succeeds.
- a || b: run b only when a fails.
- ! a: invert the success/failure result.

Three conditions with &&

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Example call: `bash check.sh report.log ERROR 10`

```
if [ "$#" -gt 2 ] &&
   [ -f "$1" ] &&
   [[ "$1" == report* ]]; then
    echo "Inputs accepted"
fi
```

1. **At least three** arguments.
2. `$1` names an **existing regular file**.
3. `$1` **starts with report**.

`[[ ... ]]` supports patterns in Bash. Keep `report*` unquoted.
&& means **AND**: stop checking as soon as one condition is false.

Checking the script's arguments

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Save as `count-lines.sh`:

```
#!/bin/bash
if [ "$#" -ne 1 ]; then
    echo "Usage: $0 FILE" >&2
    exit 2
fi
wc -l < "$1"
```

- `$#` is the argument count; `-ne 1` means **not equal to 1**.
- `>&2` sends the message to standard error.
- `exit 2` stops the script with a nonzero status.

case selects a matching pattern

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```
read -r -p "Enter a command: " choice
case "$choice" in
    all|ALL)
        ls -la ;;
    list|LIST)
        ls -l ;;
    *)
        echo "Invalid choice" ;;
esac
```

| separates alternative patterns; ;; ends a branch.
*) handles any other answer; esac ends the selection.

3. Loops and counters

How can we process several files, or ask again until the user types end?

Calculating with integers

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```
count=3
echo "$((count + 2))"
echo "$count"
```

```
5
3
```

Bash replaces `$((count + 2))` with 5 before running echo.
The calculation gives 5; `count` still contains 3.

No decimal part: `$((5 / 2))` gives 2.

Updating a counter

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```
count=3
count=$((count + 1))
echo "$count"
```

`$((count + 1))` calculates 4; the assignment stores it in `count`.

The TD also uses `expr`, which **prints** a calculation's result:

```
count=$(expr "$count" + 1)
```

`$(...)` captures that result. Here, `count` becomes 5.

for visits a list of words

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```
for group in L3info L3miage M1info; do
    echo "$group"
done
```

Each iteration assigns the next word to `group`, then runs the body.

The list can contain literal words, numbers, filenames or `"$@"`.

One iteration per script argument

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In `show-files.sh`:

```
for filename in "$@"; do
    echo "File: $filename"
done
```

Run it with:

```
bash show-files.sh notes.txt "meeting notes.txt"
```

Two iterations. `"$@"` keeps `meeting notes.txt` as one filename.

while repeats after a successful test

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```
i=1
while [ "$i" -lt 10 ]; do
    echo "$i"
    i=$((i + 1))
done
```

Test **before** each iteration.
The loop prints the integers **1 through 9**.

Asking again until the user stops

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```
read -r -p "Word (end to stop): " word
while [ "$word" != "end" ]; do
    echo "You entered: $word"
    read -r -p "Word (end to stop): " word
done
```

end makes the test false and stops the loop.
The last read updates word before the next test.

while can test a whole command list

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```
i=1
while
    echo "Checking i=$i"
    [ "$i" -le 3 ]
do
    echo "Body: $i"
    i=$((i + 1))
done
```

The **last command** before do decides whether the body runs.

The condition runs once more than the body

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```
Checking i=1
Body: 1
Checking i=2
Body: 2
Checking i=3
Body: 3
Checking i=4
```

- At **1, 2, 3**, the test succeeds.
- At **4**, the final test fails.
- The body does not run a fourth time.

4. Functions

How can we reuse a file-processing step for several files?

Defining and calling a function

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Define the function, then call it with a filename:

```
count_lines() {
    local filename="$1"
    wc -l < "$filename"
}
count_lines "names.txt"
```

- Inside the function, `$1` is **the argument passed to this call**.
- `local` gives the function its own `filename` variable.
- The result is printed on standard output, ready for a redirection or pipe.

5. Build a log report

Imagine we run a website and a night job. How many error lines did each produce?

Imagine we run two services

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Each service writes a **log**: a text file recording what happened.

Website: `web.log`

```
10:00 INFO STARTED
10:01 ERROR DISK_FULL
10:02 WARN SLOW_REQUEST
10:03 ERROR TIMEOUT
10:04 INFO REQUEST_OK
10:05 ERROR DISK_FULL
```

Night job: `worker night.log`

```
22:00 INFO STARTED
22:01 ERROR TIMEOUT
22:02 INFO JOB_DONE
22:03 WARN RETRY
22:04 ERROR QUEUE_FULL
```

How many lines contain `ERROR` in each file?

The script we want

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Let's write `report.sh` to **count matching lines in each log**.

It asks what to search for. We enter ERROR.

It saves this report in `search-report.txt`:

```
Matching lines for: ERROR
web.log: 3 matching lines
worker night.log: 2 matching lines
```

What do the arguments mean?

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```
bash report.sh web.log "worker night.log"
```

Part of the command	Role
`bash report.sh`	Run the script we will write
`web.log`	First file to read: `$1`
`"worker night.log"`	Second file to read: `$2`

The filename contains a space. Quotes keep it as **one argument**.
The script receives **two arguments**: `$#` is 2.

Let's build it together

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Download and extract the files.
Work in `report-demo` and edit `report.sh`.

```
#!/bin/bash
```

We will add and run the script **one step at a time**.

First question: what should happen if no filenames are supplied?

1. Did we receive any filenames?

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If there are none, explain how to call the script and stop:

```
#!/bin/bash
if [ "$#" -eq 0 ]; then
    echo "Usage: $0 LOGFILE..." >&2
    exit 1
fi
```

Predict: what happens with `bash report.sh` and no filename?

The script prints its usage on stderr and stops with status 1.

2. What should we search for?

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We have the filenames. Now ask the user which text to find:

```
read -r -p "Pattern to count: " pattern
echo "Searching for: $pattern"
```

Run the script with the two logs and type ERR0R.

`pattern` now contains ERR0R. We will search for it in both files.

3. Count matches in the first file

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Add this function, then pass it the first filename (`web.log`):

```
report_file() {
    local matches
    matches=$(grep "$pattern" "$1" | wc -l)
    echo "$1: $matches matching lines"
}
report_file "$1"
```

`grep` selects the lines; `wc -l` counts them.
For ERROR, this call prints `web.log: 3 matching lines`.

4. Repeat for every filename

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Replace `report_file "$1"` with a loop over **all the filenames**:

```
for logfile in "$@"; do
    if [ -f "$logfile" ] && [ -r "$logfile" ]; then
        report_file "$logfile"
    else
        echo "Cannot read file: $logfile" >&2
        exit 1
    fi
done
```

Predict: does `worker night.log` cause one iteration or two?

5. Save the counts in a report

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Create `search-report.txt` **once, before the loop**:

```
report="search-report.txt"
echo "Matching lines for: $pattern" > "$report"
```

Inside the loop, **append each file's count** to that report:

```
report_file "$logfile" >> "$report"
```

Why >> here? Each file's result must be added after the previous one.

After the loop: `echo "Report saved in $report"`.

6. Do we get the expected report?

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```
bash report.sh web.log "worker night.log"
```

Type ERROR, then run `cat search-report.txt`:

```
Matching lines for: ERROR
web.log: 3 matching lines
worker night.log: 2 matching lines
```

Try WARN, FATAL, a missing file and a call with no arguments.
Complete script

CM3 summary: writing a script

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1. Get inputs with `read` or the script's arguments.
2. Store values, quote them and check the inputs.
3. Use tests and loops to decide what to repeat.
4. Put a reusable task in a function.
5. Run the script with normal and incorrect inputs; inspect its output.

Further examples and TP tools

Reference for the TD/TP: extra Bash forms, worked examples and text-editing tools

Options for the TP script

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```
bash main.sh -p samples -h
```

Bash passes these as ordinary arguments. **The script interprets them** with `getopts "hp:" option`, reading one option at a time:

Item	Meaning
`h`	Accept `-h` without a value
`p:`	Accept `-p` with a required value
`option`	Variable receiving the option letter
`OPTARG`	Variable receiving its value, here `samples`

Reading options with getopts

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```
directory="."
while getopts "hp:" option; do
    case "$option" in
        h) echo "Usage: $0 [-p DIRECTORY]"; exit 0 ;;
        p) directory="$OPTARG" ;;
        *) exit 1 ;;
    esac
done
echo "Selected directory: $directory"
```

The loop runs while another option is available.
An unknown option or a missing value reaches the `*)` branch.

Removing arguments already handled

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After `bash inspect.sh -p logs app.log`, the script can do:

```
directory="$2"
shift 2
echo "$directory"
echo "$1"
```

Output: `logs`, then `app.log`.
`shift 2` removes the first two arguments and renumbers the remaining ones.

After `getopts`, `OPTIND` is the position of the next argument.
`shift "$((OPTIND - 1))"` removes the arguments it has read.

Replacing text with sed

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For a file containing `cat cat`:

```
sed 's/cat/dog/' animals.txt
sed 's/cat/dog/g' animals.txt
```

Output: `dog cat`, then `dog dog`.

- `s/old/new/` replaces the first match **on each line**.
- The final `g` replaces every match on each line.
- These commands print the result and leave the file unchanged.

`sed -i 's/cat/dog/g' animals.txt` changes the file itself.
Practise on copies.

Bash: Bourne again

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Bash = Bourne-Again SHell.

The name puns on **Stephen Bourne**, author of the earlier Unix shell `sh`.

We use Bash syntax in this course.
`sh script.sh` may select a different shell with fewer features.

The shebang: selecting an interpreter

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```
#!/bin/bash
printf 'Hello, MIAGE\n'
```

On Linux, direct execution of `./hello.sh` uses the interpreter named after `#!`.

The extension `.sh` does not choose Bash.
The first line does. We call it the **shebang**.

One symbol, several names

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Symbol	Names you may hear
#	hash, number sign; pound sign in US usage
#	French: **croisillon**, also **carré**
♯	The musical sharp sign (*dièse*)

#!: **hash** followed by **bang**.
The symbol's name and its role depend on context.

Quoted variable expansions

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```
filename="meeting notes.txt"
cat "$filename"
echo '${filename}'
echo "${filename}"
```

- Double quotes preserve the expanded value as one argument.
- Single quotes preserve the literal text `${filename}`.

Printing values

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```
name="Alice"
echo "Hello, $name"
printf 'Hello, %s\n' "$name"
```

echo is convenient for simple messages.
`printf` gives explicit control over formatting and newlines.

A child receives a copy

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```
export message="parent"
bash -c 'message="child"; echo "$message"'
echo "$message"
```

```
child
parent
```

The condition can be grep

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```
if grep ' ERROR ' app.log; then
    echo "At least one error was found"
else
    echo "No match, or grep could not read the file"
fi
```

`if` runs `then` when the command returns status **0**.
Any other status selects `else`. Here, `grep` succeeds when it finds a matching line.

Which messages appear?

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```
true  || echo "A"
false || echo "B"
true  && echo "C"
false && echo "D"
```

```
B
C
```

Checkpoint: command status

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1. Status `0`: true or false?
2. Does `false; echo yes` print?
3. Does `false && echo yes` print?
4. Status of `false || true`?

1. True / success.
2. Yes. `;` always continues.
3. No.
4. `0`, from the final executed command, `true`.

Read a file line by line

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```
while IFS= read -r name; do
    printf 'Hello, %s\n' "$name"
done < names.txt
```

- `IFS=` disables trimming of leading/trailing spaces for this `read`.
- `-r` keeps backslashes unchanged, as in the earlier prompt.
- `read` succeeds for each complete line; at **EOF**, it fails and the loop stops.
- `< names.txt` supplies input to the **whole loop**.

until repeats after a failed test

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```
i=1
until [ "$i" -ge 10 ]; do
    echo "$i"
    i=$((i + 1))
done
```

This also prints **1 through 9**.
The loop stops when the condition succeeds.

break, continue and exit

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Command	Effect
`continue`	Skip the rest of the current iteration
`break`	Leave the current loop
`exit n`	End the script with status `n`

After `break`, execution continues **after the loop**.

Command substitution

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```
echo "Current directory: $(pwd)"
```

The shell runs pwd and replaces `$(pwd)` with its **standard output**.

The older form is `` `pwd` ``.
We will write `$(pwd)` because nested commands are easier to read.

Newlines, semicolons and continuations

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```
echo "first"
echo "second"; echo "third"
echo "a longer message" \
     "continued on the next line"
```

- A newline or ; separates commands.
- A backslash immediately before a newline continues the line.

Integer, readonly and array variables

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```
declare -i count=3
readonly course="NUnix"
declare -a words
words[0]="one"
words[1]="two"
echo "${words[0]}"
```

`unset count` removes a variable. A readonly variable cannot be reassigned or unset in that shell.

Familiar environment variables

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Variable	Meaning
`HOME`	Home directory
`USER`	Account name, usually set at login
`PWD`, `OLDPWD`	Current and previous working directories
`PATH`	Colon-separated command search directories
`SHELL`	Configured login shell, not necessarily the current shell
`TERM`	Terminal type
`HOSTNAME`	Host name in Bash
`MANPATH`	Manual search path, when explicitly set

Arithmetic with let and expr

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```
let 'count = 3 + 5'
echo "$count"
expr 3 + 5
expr 3 \* 5
```

- `let` evaluates an arithmetic expression in Bash.
- `expr` takes separate arguments, so its spaces matter.
- Escape * so the shell passes it to `expr` literally.

Bash arithmetic shortcut: ((...))

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```
count=3
((count = count + 1))
echo "$count"
```

```
4
```

`((count = count + 1))` tells Bash to calculate and update `count`. It prints nothing; `echo` displays the new value.

Here, it does the same update as `count=$((count + 1))`.

Bash tests with double brackets

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```
filename="report.txt"
if [[ "$filename" == *.txt ]]; then
    echo "Text file"
fi
```

`[[ ... ]]` is another form of test available in Bash.
Here, `== *.txt` checks whether the name **ends in `.txt`**.

With `[ ... ]`, string equality compares literal text; this Bash form also accepts patterns.

Spaces are still required: `[[ condition ]]`.

Checkpoint: arithmetic and tests

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1. Value of `$((5 / 2))`?
2. Is `n = 3` an assignment?
3. Status of `false`?
4. Does `-f` accept every existing directory?

1. 2 (integer division).
2. No. Use n=3.
3. 1
4. No. It tests for a regular file.

Checking access to a file

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```
#!/bin/bash
read -r -p "Enter a filename: " filename
if [ ! -w "$filename" ]; then
    echo "File is not writable"
    exit 1
elif [ ! -r "$filename" ]; then
    echo "File is not readable"
    exit 1
else
    echo "File is writable and readable"
fi
```

! negates a test. `-w` checks writing; `-r` checks reading.

Comparing two files

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```
diff "$1" "$2" > diff.out
status=$?
if [ "$status" -eq 0 ]; then
    echo "Files are identical"
elif [ "$status" -eq 1 ]; then
    cat diff.out
else
    echo "Comparison failed" >&2
fi
rm -f diff.out
```

`status=$?` saves the previous command's status. For `diff`:
0: identical; **1**: different; **greater than 1**: error.

Copying a known list of files

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```
#!/bin/bash
destination=$1
for file in f1 f2 f3; do
    cp "$file" "$destination/$file"
done
```

Call with an **existing destination directory** as the first argument.

A loop can feed a pipeline

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```
for number in 6 3 1 2; do
    echo "$number"
done | sort -n
```

```
1
2
3
6
```

Repeating until interrupted

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```
while true; do
    echo "alive..."
    sleep 3
done
```

`true` always succeeds.
Use **Ctrl+C** to interrupt this foreground loop.

Skipping 4, stopping at 7

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```
for x in 1 2 3 4 5 6 7 8 9; do
    if [ "$x" -eq 4 ]; then
        continue
    fi
    echo "$x"
    if [ "$x" -eq 7 ]; then
        break
    fi
done
echo "After the loop: x=$x"
```

The loop's output

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```
1
2
3
5
6
7
After the loop: x=7
```

- At **4**, `continue` skips `echo`.
- At **7**, `echo` runs before `break`.
- The final message runs after the loop.

Function scope

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```
label="outside"
show_label() {
    local label="inside"
    echo "$label"
}
show_label
echo "$label"
```

```
inside
outside
```

The local `label` hides the outer variable during the call.
The outer value is available again when the function finishes.

Builtins: shell and process control

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Builtin	Use
`cd`, `pwd`	Change / report the current directory
`history`	Display command history
`exec`	Replace the shell with another program
`exit`	End the current shell
`kill`	Send a signal
`trap`	Arrange a response to a signal or shell event

Builtins: data and parameters

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Builtin	Use
`declare`, `readonly`, `local`	Define variable attributes or scope
`export`, `set`	Manage the environment / shell settings
`echo`, `read`	Print output / read input
`let`, `test`	Arithmetic / condition tests
`shift`	Discard leading positional parameters
`return`	End a function and report its status

Reading a small script

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```
#!/bin/bash
for filename in "$@"; do
    if [ -f "$filename" ]; then
        wc -l "$filename"
    fi
done
```

For each argument that names a regular file, print its line count.

Optional: bash -n and -x

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```
bash -n count-lines.sh
bash -x count-lines.sh notes.txt
```

- `-n`: check the syntax without running the script.
- `-x`: run the script and print each expanded command on stderr.